

AI Usage & Personal Contribution Statement

Heart Disease Prediction – IITM BS MLOps OPPE-2 (May 2026)

Purpose of this document

This document records how AI tools were used during the project. AI was used as a **development assistant, tutor, debugging partner, documentation assistant and review tool**, rather than as a replacement for the student's work. The student performed the actual execution, configuration, testing, verification and deployment activities in the project environment.

How AI was used

Area	AI assistance	Student's hands-on contribution
ML model development	Explained preprocessing, train/test splitting, imputation, scaling, loading the data, splitting, regression	loaded the data, split the data into training/missing values, executed the train
Debugging	Helped interpret Python, scikit-learn and environment errors	As suggested, corrected the kernel when required, verified package
SHAP explainability	Explained SHAP concepts and helped structure an importance	Executed SHAP analysis, interpreted the feature-importance values and inc
Fairness	Explained Fairlearn metrics and how to interpret demographic	Used Fairlearn to identify sensitive attribute, executed the Fairlearn analy
FastAPI	Suggested API structure, request schema and prediction-resp	Created and tested the API, validated requests, ran Uvicorn and verified t
Docker	Provided Dockerfile structure and troubleshooting guidance.	Built the image, resolved a local port conflict, ran the container and tested
GCP / Artifact Registry	Explained registry authentication, image tagging and permissi	Configured Docker authentication, tagged the image, handled IAM permis
GKE / Kubernetes	Provided Kubernetes manifest examples and explained Deplo	Created and applied the Deployment, LoadBalancer Service and HPA, verified
CI/CD	Helped construct and debug the GitHub Actions workflow and	Created the workflow, configured GitHub/GCP Workload Identity Federati
Observability	Suggested structured prediction logging and explained how	Implemented structured logging, generated 100 individual API predictions
Performance testing	Explained how to use wrk and how to interpret throughput, l	Installed wrk, executed the required 2,000-concurrent-connection test and
Drift monitoring	Explained PSI and provided a method for comparing training	Generated the PSI on the prediction dataset, calculated PSI values and inte

Examples of AI-assisted problem solving

- **Environment issue:** When scikit-learn imports failed because of an incompatible environment, AI helped identify the likely package-version problem and suggested a compatible scikit-learn version. The student executed the installation and verified the corrected environment.
- **GitHub Actions issue:** AI helped identify malformed YAML indentation and later interpreted the GitHub OIDC error. The student corrected the workflow, added the required OIDC permission and pushed the changes.
- **GCP IAM issue:** AI interpreted permission-denied messages and helped choose Workload Identity Federation when service-account key creation was blocked by organization policy. The student configured the existing GCP resources through the console.
- **Deployment issue:** AI explained the meaning of the Kubernetes rollout state and image-tag problem. The student executed the kubectl commands and verified the rollout.
- **Performance result:** AI helped interpret the wrk output. The student actually ran the 30-second, 2,000-connection test and obtained the measured metrics.

What was not delegated to AI

- AI did not independently access or operate the student's GCP/GitHub accounts.
- AI did not independently create or execute the student's cloud resources.

- AI did not independently submit the OPPE or represent that testing had occurred without the student's execution.
- Commands and configuration changes were entered and verified by the student in the actual project environment.
- Observed metrics and outputs were taken from the student's environment rather than invented for the report.

Project evidence produced through hands-on execution

Evidence	Result
Trained model	heart_disease_model.joblib
API	FastAPI /predict and /health endpoints tested successfully
Container	Docker image built and tested locally
Cloud deployment	GKE Deployment + LoadBalancer Service
Autoscaling	HPA configured with maximum 3 replicas
CI/CD	GitHub Actions workflow successfully authenticated through OIDC and built/pushed the image
100-sample prediction run	100 / 100 successful API predictions
Load test	2,000 concurrent connections; 911.67 requests/sec; 886.85 ms average latency
Drift analysis	PSI calculated for the generated prediction dataset versus training data

Summary: AI was used throughout the project as a technical support tool for explanation, code suggestions, debugging, troubleshooting and documentation. The student remained responsible for implementing the solution in the actual environment, making configuration decisions, executing commands, testing the deployed system and validating the resulting outputs.